Lab 53: Taxonomy drift: write it to a state file

Topic 5: Context Management & Reliability · Task 5.1: Manage conversation context to preserve critical information across long interactions

Runs on Simulator 3: Multi-Agent Research System · Self-practice

Source: SkillCertPro practice exam Part 1, Q21 (unofficial third-party question)

Commit to an answer first

During a long research run, early document analysis identifies a domain-specific taxonomy that should drive the final report. After many searches, tool results, and revisions, later synthesis passes start organizing findings around generic categories and occasionally contradict the earlier taxonomy, even though the application resends the accumulated message history. What change would best improve reliability?

A. Maintain a structured research-state file with the discovered taxonomy and source mappings, then load it into later synthesis prompts.

B. Enable prompt caching for the accumulated prefix so earlier findings can be reused without affecting the context window.

C. Add a synthesis instruction to avoid generic categories unless the full transcript explicitly supports them.

D. Increase max_tokens for later synthesis calls and continue sending the complete accumulated transcript unchanged.

Simulate the solution

```
# In the browser: open Simulator 3, then Lab 53
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 53

# In a terminal, from simulators/simulator-3-multi-agent-research/
uv run research_system.py --case UC-21
```

What you should see

research_state.json holds taxonomy and source map, and the synthesis packet quotes it.

`evidence/research_state.json` exists with a taxonomy and source map; the synthesis packet quotes it.

Open `simulators/simulator-3-multi-agent-research/evidence/` and find it yourself.

The answer

A. A structured research-state file loaded into later prompts.

Having information somewhere in the request is not the same as making it usable at synthesis time. More context is not automatically better, because accuracy and recall degrade as tokens grow. A structured state file (scratchpad) keeps the taxonomy, key claims and source mappings compact and explicit, so the coordinator can place them where later passes reliably use them. The Messages API is stateless, so the application assembles the relevant context on every request.

Why the others fail

B. Rests on a false premise: cached tokens still count toward the context window, and caching does not restore attention to early findings.

C. Relies on the model finding the buried taxonomy again inside a long transcript.

D. max_tokens caps output. Sending the transcript unchanged keeps degrading recall.

Check yourself

- save_research_state called before load_research_state
- research_state.json holds a taxonomy with >=3 sectors and a source map
- synthesis packet contains the taxonomy sector names

Now change one thing

In fixtures/sources.json replace the visual-arts TIMEOUT with [] and rerun: a valid empty result must not read as a failure.

Third-party practice material from SkillCertPro, reconstructed from a video transcript in the Tertiary Markdown Vault. Not a real exam item. Wording may differ from the original.